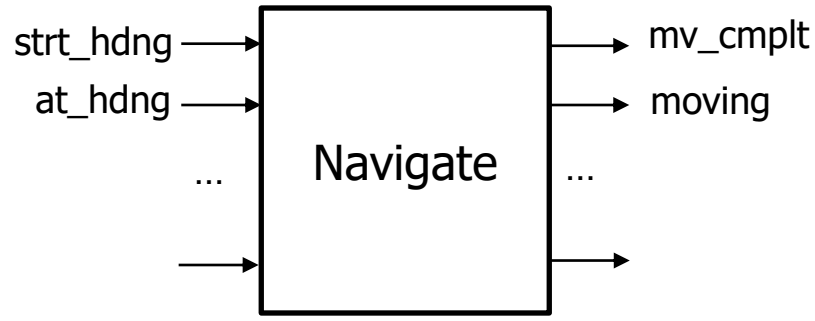
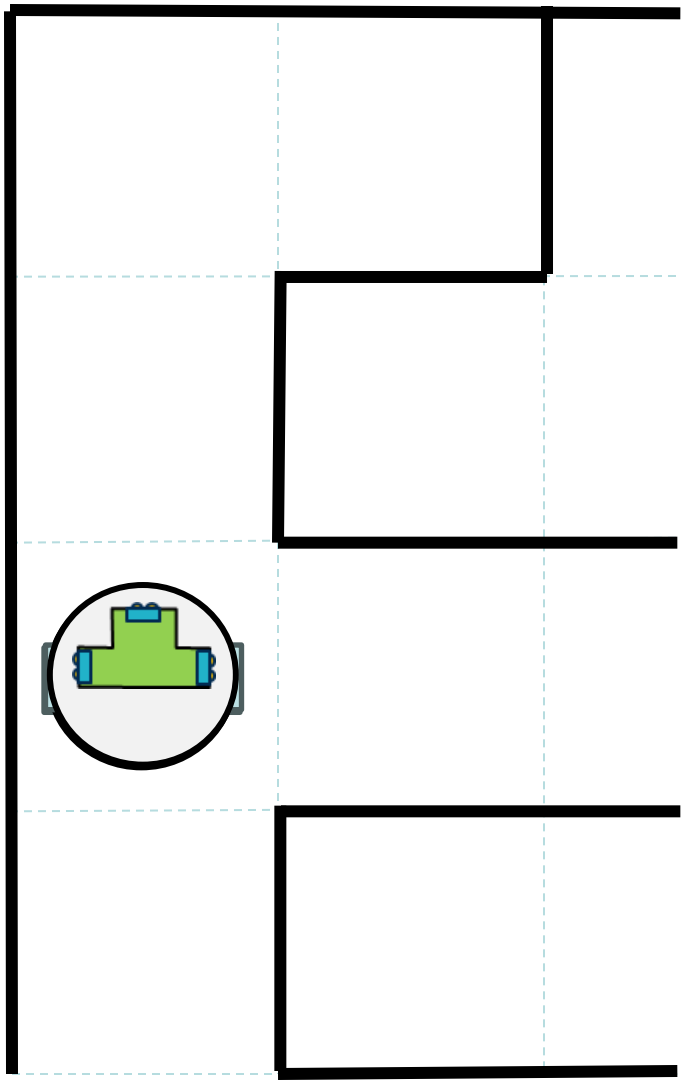
Exercise 13: Navigate Unit

- Receives commandand decide how robot will move
- 2 types of commands:
  - Change heading: **strt_hdng**
  - Straight line movement: **strt_mv**
- Change heading
  - Get **strt_hdng** cmd
  - Assert **moving** until **at_hdng** asserted
  - Assert **mv_cmplt**





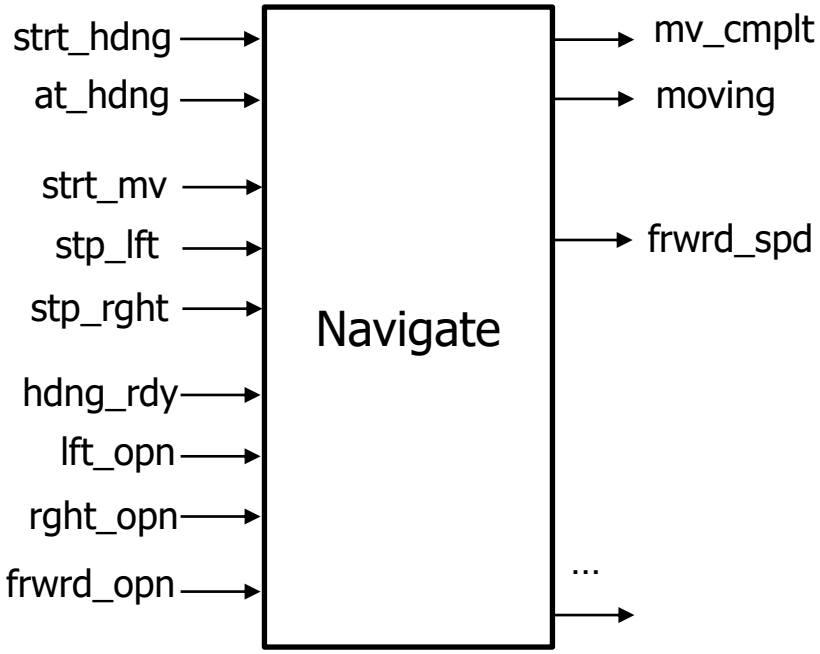
Exercise 13: Navigate Unit

Move command

- Get **strt_mv** cmd
- Accelerate **frwrd_spd** to MAX_FRWRD
  - Set initial speed by asserting **init_frwrd**
  - Assert **moving**
  - Ramp up frwrd_spd by asserting **inc_frwrd**
- On stopping condition, decelerate **frwrd_spd** to 0

Stopping Conditions

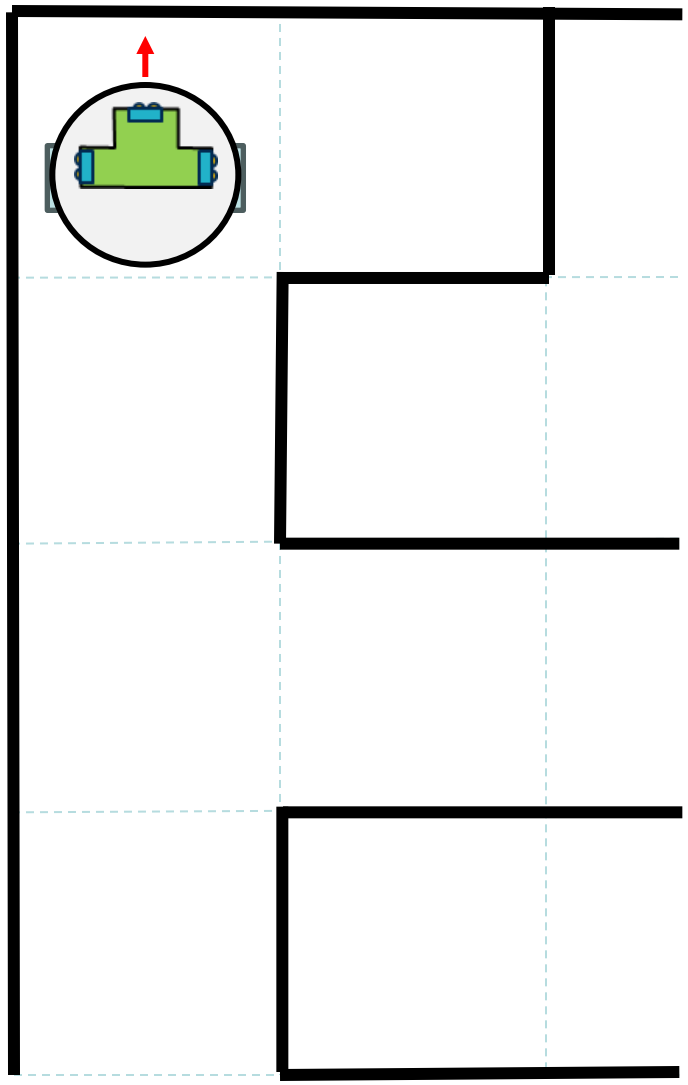
- We encounter a left opening (**stp_lft** cmd, lft_opn)
- We encounter a right opening (**stp_rght** cmd, rght_opn)
- We encounter wall in front (~frwrd_opn)



Exercise 13: Navigate Unit

We need 2 types of deceleration

- If we are at a wall, need to decelerate fast so we don't crash → decel. at 8x normal incr.

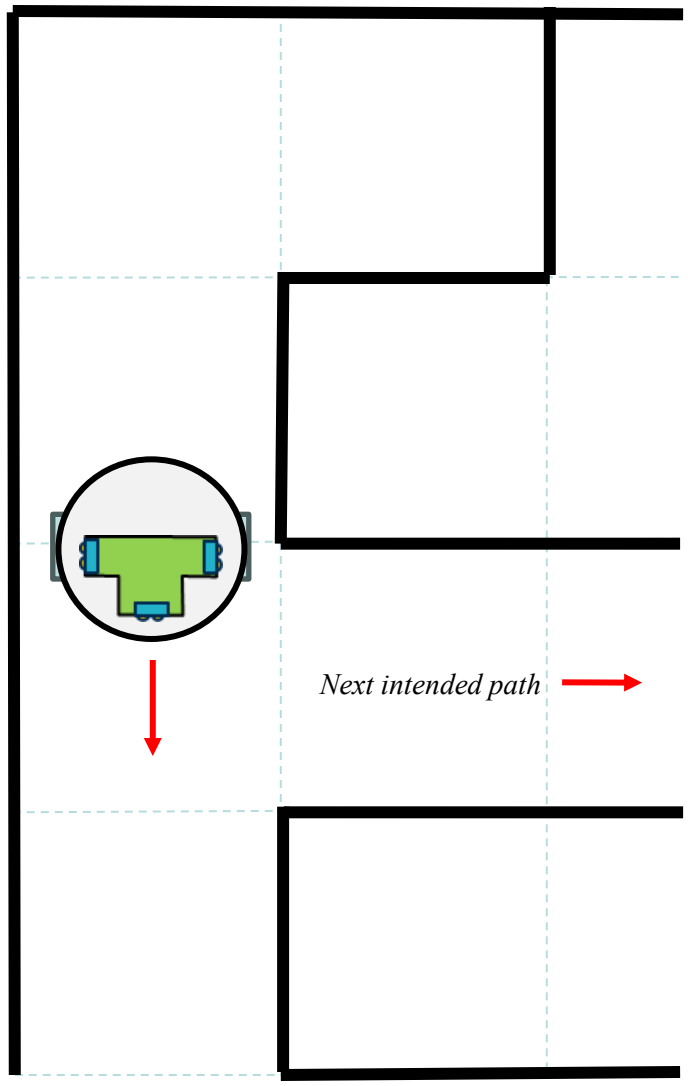


Exercise 13: Navigate Unit

We need 2 types of deceleration

- If we are at a wall, need to decelerate fast so we don't crash
  → decel. at 8x normal incr.

- If we are at a requested opening, need to decelerate so we travel another ½ square
  → decel. at 2x normal incr.



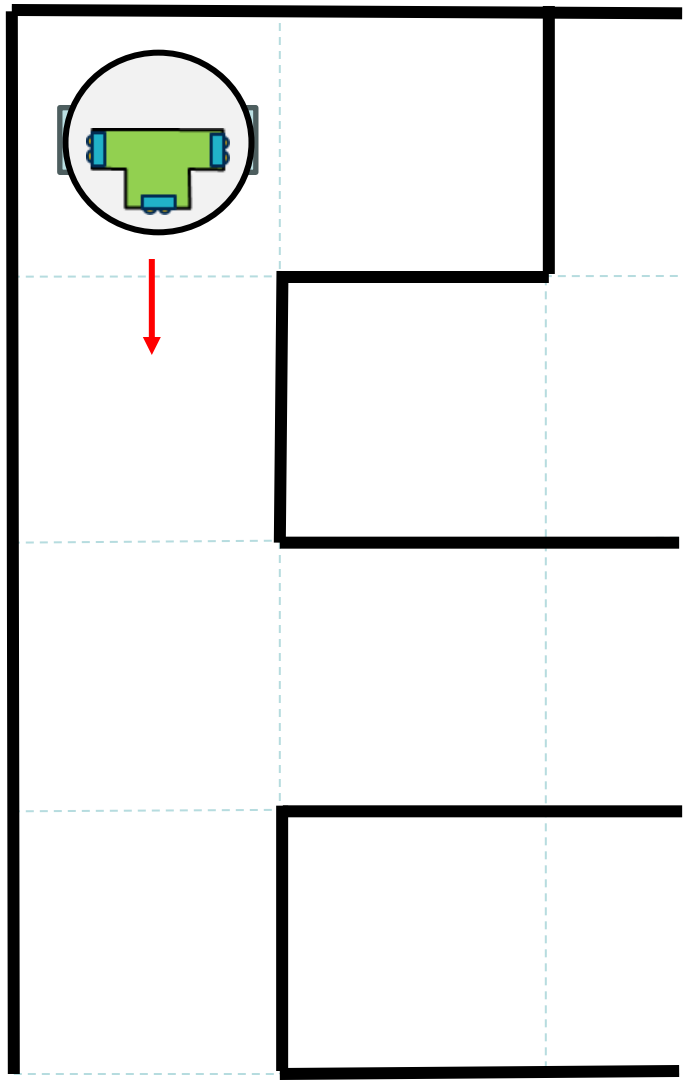
Exercise 13: Navigate Unit

We need 2 types of deceleration

- If we are at a wall, need to decelerate fast so we don't crash
  → decel. at 8x normal incr.

- If we are at a requested opening, need to decelerate so we travel another ½ square
  → decel. at 2x normal incr.

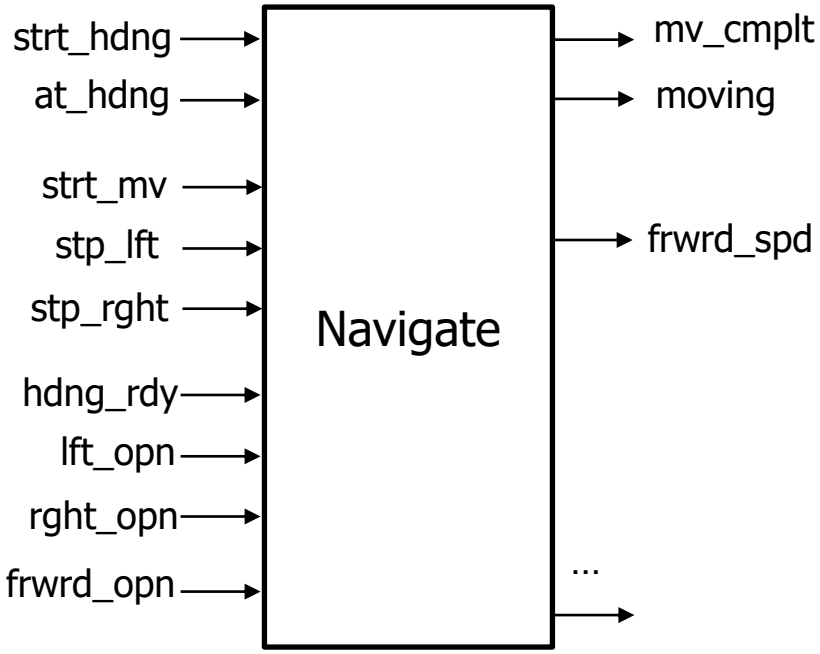
Note:
In order to start moving when requested opening is already asserted (e.g., lft_opn), you need **rising edge detectors**
→ lft_opn_rise and rght_opn_rise



Exercise 13: Navigate Unit

Move command

- Get **strt_mv** cmd
- Accelerate **frwrd_spd** to MAX_FRWRD
  - Set initial speed by asserting **init_frwrd**
  - Assert **moving**
  - Ramp up frwrd_spd by asserting **inc_frwrd**
- On stopping condition, decelerate **frwrd_spd** to 0
  - **Fast decel**: ramp frwrd_spd down fast by asserting **dec_frwrd_fast**
  - **Normal decel**: ramp frwrd_spd down by asserting **dec_frwrd**
  - Assert **mv_cmplt**



Exercise 13: Navigate Unit

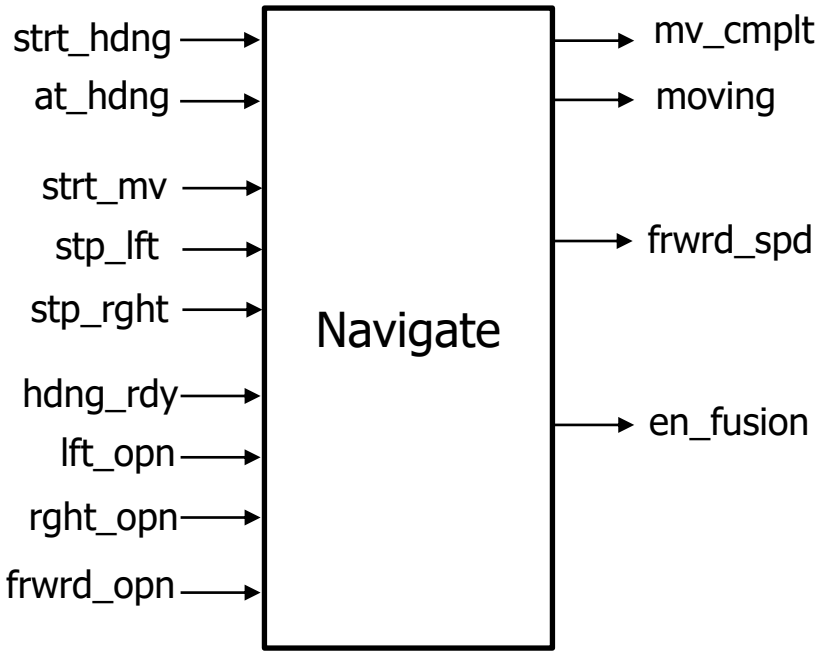
Generate **en_fusion**

- We want the IR sensors to affect navigation ONLY if it is moving forward
- Decode the current magnitude of frwrd_spd to assert en_fusion if it is greater than ½ MAX_FRWRD

Implement parameter **FAST_SIM**

- Rate of incr. / decr. is controlled by magnitude of **frwrd_inc**
- Normal operation: 6'h02
- This will take long to simulate, use larger step during simulation
- Simulation: 6'h18

```
generate if (FAST_SIM) begin
  // set frwrd_inc to 6'h18
end else begin
  // set frwrd_inc to 6'h02
end
endgenerate
```



Exercise 13: Navigate Unit

- You are provided the **frwrd_spd** register implementation in **navigate_shell.sv** → Use this to create your **navigate.sv**
  - Add edge detectors on **lft_opn** and **rght_opn**
  - Implement frwrd_inc based on parameter FAST_SIM
  - Implement en_fusion signal
  - Implement the FSM
    - Understand requirements
    - Declare states and SM outputs
    - Implement state registers
    - Implement state transition and output logic

- A testbench with a few test cases is provided (**navigate_tb_starter.sv**)
- Complete it with more test cases & submit **navigate_tb.sv**
- Submit proof that it passed the self-checking testbench (transcript window)

Navigate Unit: FSM Functionality

- Sit in IDLE till told to turn to a new heading (**strt_hdng** asserted) or start a move (**strt_mv** asserted).
- If performing a heading change, simply assert **moving** and wait until **at_hdng** signal *(from **PID** unit)* is asserted. Assert **mv_cmplt** on way back to IDLE.
- If performing a move first initialize forward speed (**init_frwrd**) and then ramp up to speed. In this state assert **moving** and **inc_frwrd**.
  - Stay in the accelerate state because the **frwrd_spd** register will saturate to **MAX_FRWRD** on its own. So effectively this is both accelerate and "cruise" state.
  - Leave the accelerate state when IR readings indicate stopping condition *(either obstacle in front (~frwrd_opn) or desired opening lft/right)*
- There are 2 decelerate states, they differ only in whether one is asserting **dec_frwrd** or **dec_frwrd_fast**. The SM stays in these state until the **frwrd_spd** register is 0. Even though we are in the process of stopping in these states, **moving** should still be asserted. These states return to IDLE when completed *(and assert **mv_cmplt** on the way)*

SM output signals
moving
init_frwrd
inc_frwrd
dec_frwrd
dec_frwrd_fast
mv_cmplt